

VƯƠNG VIỆT HÙNG

PERSONAL DETAILS

Full name:	Vương Việt Hùng	Nationality:	Vietnamese
Phone:	+84-583-885-391	Languages:	Vietnamese (Native)
Email:	vuongviethung156@gmail.com		English (IELTS 7.5)
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EDUCATIONAL QUALIFICATIONS AND ACADEMIC AWARDS

Soongsil University	Seoul, South Korea
Master's Degree in Electronic Engineering	March 2024 – March 2026
– School of Electronic Engineering	
– GPA: 4.24/4.50	
– Thesis title: Online Continual Learning for LiDAR Panoptic Tracking	
– Thesis length: 10,387 words	
Hanoi University of Science and Technology	Hanoi, Vietnam
Bachelor's Degree in Control Engineering and Automation	August 2019 – September 2023
– School of Electronic and Electrical Engineering	
– GPA: 3.34/4.00	
– Thesis title: Application of Stereo Matching in SLAM for Autonomous Vehicles	
– Thesis length: 13,536 words	

PROFESSIONAL AFFILIATIONS AND MEMBERSHIPS

None to date

EMPLOYMENT HISTORY

VinDynamics	Hanoi, Vietnam
SLAM & Navigation Engineer	March 2026 – current
– Worked on visual-inertial odometry and multi-sensor height mapping for real-time perception on humanoid robotic platforms.	
– Contributed to VIO pipeline profiling and latency optimization, and extended GPU-based height mapping with LiDAR, depth camera, IMU, and joint-state integration.	
ANDA Lab, Soongsil University	Seoul, South Korea
Research Assistant	March 2024 – March 2026
– Conducted research on LiDAR panoptic segmentation and tracking, with a focus on domain adaptation and online continual learning.	
– Contributed to model development and evaluation, research publications, researcher mentoring, and technical seminars.	
Rikai Mind	Hanoi, Vietnam
AI Engineer Intern	August 2022 – January 2023

- Developed a computer vision system for object volume estimation from monocular images and camera–object distance, using CNN-based features and keypoint regression.
- Contributed to data collection, model development and evaluation, and deployment of the inference pipeline through a FastAPI-based service.

OTHER RESEARCH EXPERIENCE

Mandevices Laboratory, Hanoi University of Science and Technology

Hanoi, Vietnam

Research Assistant (Voluntary)

August 2020 – September 2023

- Conducted research on deep-learning-based bearing fault diagnosis, including simulation-based domain adaptation to address limited real-world training data.
- Contributed to data collection, model development and evaluation, research publications, and researcher mentoring.

RESEARCH OUTPUTS

Refereed journal articles

1. **Vuong, H. V., & Yoo, M.** (2026). Domain adaptive LiDAR panoptic segmentation via pseudo-supervised instance learning. *IEEE Transactions on Intelligent Transportation Systems*. <https://doi.org/10.1109/TITS.2026.3695516>
Journal Impact Factor: 9.1 (2025, JCR)
 - Contribution: Designed and implemented the proposed methodology, designed and conducted the experiments, analyzed and interpreted the results, and prepared the manuscript.
 - Research outcome: Proposed PILLA for instance-level pseudo-supervised domain adaptation, achieving 35.97% PQ on nuScenes and outperforming AdaptLPS by 13.14 percentage points.
2. Ha-Phan, N.-Q., **Vuong, H. V., & Yoo, M.** (2025). Instance embedding as queries for DETR-based LiDAR panoptic segmentation. *IEEE Transactions on Intelligent Vehicles*, 10(10), 4629–4641. <https://doi.org/10.1109/TIV.2024.3488035>
Co-first authors: Ha-Phan, N.-Q. and Vuong, H. V.
Journal Impact Factor: 14.3 (2025, JCR)
 - Contribution: Primarily designed and implemented the proposed methodology, and jointly conducted the experiments, analyzed and interpreted the results, and prepared the manuscript with the co-first author.
 - Research outcome: Proposed a DETR-based architecture using instance embeddings as object queries, achieving 63.40% PQ on SemanticKITTI and 77.08% PQ on nuScenes, outperforming previous methods by 1.20 and 1.94 percentage points, respectively.
3. Nguyen, K. N. M., **Vuong, H. V.,** Ha-Phan, N.-Q., & Yoo, M. (2025). FuPaSCo: Long-range and local context fusion for 3D panoptic scene completion. *Image and Vision Computing*, 163, 105776. <https://doi.org/10.1016/j.imavis.2025.105776>
Journal Impact Factor: 5.0 (2025, JCR)
 - Contribution: Provided consultation on the methodology, assisted with the experiments, and developed an evaluation tool for performance analysis. Reviewed the manuscript and provided feedback during its preparation.

- Research outcome: Developed FuPaSCo, a dual-branch CNN–Transformer framework for LiDAR panoptic scene completion, achieving 26.10% PQ[†] on SemanticKITTI and outperforming PaSCo by 2.49 percentage points.
4. Nguyen, T. H., **Vuong, V. H.**, Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). Generalized simulation-based domain adaptation approach for intelligent bearing fault diagnosis. *Arabian Journal for Science and Engineering*, 49(12), 16941–16957. <https://doi.org/10.1007/s13369-024-09282-1>
Journal Impact Factor: 3.1 (2025, JCR)
- Contribution: Assisted with data collection, developed the bearing vibration signal model, and conducted experiments combining the signal model with the baseline deep-learning approach. Contributed to the analysis and interpretation of the experimental results.
 - Research outcome: Developed a simulation-based domain adaptation approach using synthetic bearing vibration signals, achieving 74–98% fault classification accuracy across different bearing types.

Published conference proceedings

1. Nguyen, T. H., Le, H. A., **Vuong, V. H.**, Tran, L. D. A., Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). ConvMamba: A data-efficient neural network for bearing fault diagnosis. In *2024 7th International Conference on Green Technology and Sustainable Development (GTSD)* (pp. 165–172). IEEE. <https://doi.org/10.1109/GTSD62346.2024.10674880>
- Contribution: Designed and implemented the proposed methodology, designed and conducted the experiments, and led the analysis and interpretation of the results. Led the preparation of the manuscript with assistance from the co-authors.
 - Research outcome: Developed ConvMamba, a data-efficient architecture combining convolutional networks and Vision Mamba, improving classification accuracy by 27 percentage points over Hybrid-CNN-MLP on the CWRU bearing fault dataset.

Manuscripts under review

1. Pham, T. P., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). SM3-PSC: Sparse multi-modal Mamba for 3D panoptic scene completion with multi-scale feature refinement. *Manuscript submitted to IEEE Transactions on Image Processing*.
- Contribution: Provided consultation on the proposed methodology and reviewed the manuscript, providing technical feedback and suggestions during its preparation.
 - Research outcome: Developed SM3-PSC, a sparse multi-modal framework combining RGB and LiDAR features with Mamba-based context modeling, achieving 28.20% PQ[†] on SemanticKITTI while running the 3D pipeline 2.8× faster than FuPaSCo.
2. Nguyen, K. N. M., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). LW-PSC: Lightweight design for LiDAR panoptic scene completion. *Manuscript submitted to Applied Soft Computing*.
- Contribution: Provided consultation on the proposed methodology and reviewed the manuscript, providing technical feedback and suggestions during its preparation.
 - Research outcome: Developed LW-PSC, a lightweight BEV-based framework for LiDAR panoptic scene completion, achieving 13.59% PQ at 11.3 FPS while reducing the model size from 40.65M to 29.16M parameters.

RESEARCH GRANTS AND RELEVANT AWARDS

None to date